Introduction into Possibilistic Logic

Possibilistic Logic - An Overview by Didier Dubois and Henri Prade

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Overview

Possibilistic logic

Possibilistic logic is a weighted logic that handles uncertainty in a qualitative way by adding certainty to classical logic formulas.

- **Applications**: Useable in expert systems, decision-making and for reasoning under uncertainty.

Overview

Possibility Distribution

Possibility distribution

A **possibility distribution** is a mapping:

$$\pi : U \to S,$$

where:

- U: Set of states
- S: Totally ordered scale ($S = [0, 1]$ or finite chains).
- $\pi(u)$: Degree of possibility (plausibility) of state u.

Example

- $\pi(u) = 0$: State u is **impossible**.
- $\pi(u) = 1$: State u is **fully possible**.

Possibility measure

$$\Pi(A) = \sup_{u \in A} \pi(u).$$

Evaluates the extent to which A is consistent with π.

Necessity measure

$$N(A) = \inf_{u \notin A} (1 - \pi(u)).$$

Evaluates the extent to which A is certainly implied by π.

Overview

Basic possibilistic logic formula

A basic possibilistic logic formula is represented as:

$$(a, \alpha)$$

where:

- a: A classical logic formula (e.g., $p \rightarrow q$).
- $\alpha \in (0, 1]$: Certainty level (minimum necessity degree of a).

Example

- $(b \rightarrow f, 0.9)$: Birds fly with certainty 0.9.
- $(p \rightarrow \neg f, 1)$: Penguins do not fly, absolutely certain.

Certainty weakening

$$\text{If } \beta \leq \alpha, \quad (a, \alpha) \vdash (a, \beta)$$

Example

(It will rain tomorrow, 0.9) $\vdash$ (It will rain tomorrow, 0.7)
If the certainty level decreases, the conclusion still holds.

Modus ponens

$$(\neg a \vee b, \alpha), (a, \alpha) \vdash (b, \alpha), \quad \forall \alpha \in (0, 1]$$

Example

$(\neg \text{fever} \vee \text{flu}, 0.8), (\text{fever}, 0.8) \vdash (\text{flu}, 0.8)$
If fever is observed, flu is inferred with certainty 0.8.

Resolution

$$(\neg a \vee b, \alpha), \quad (a \vee c, \alpha) \vdash (b \vee c, \alpha)$$

Example

- Propositions:
  - $a =$ "It is raining",
  - $b =$ "The ground is wet",
  - $c =$ "There is a thunderstorm".
- Resolution:

$$(\neg a \vee b, 0.7), (a \vee c, 0.7) \vdash (b \vee c, 0.7)$$

- Interpretation: "The ground is wet or there is a thunderstorm" holds with certainty 0.7.

Weakest link resolution

$$(\neg a \vee b, \alpha), \quad (a \vee c, \beta) \vdash (b \vee c, \min(\alpha, \beta))$$

Example

- Premises:

$$(\neg a \vee b, 0.7), \quad (a \vee c, 0.5).$$

- Inference:

$$(\neg a \vee b, 0.7), \quad (a \vee c, 0.5) \vdash (b \vee c, \min(0.7, 0.5)).$$

- Interpretation: "The ground is wet or there is a thunderstorm" is true with certainty 0.5, limited by the weaker premise.

Knowledge base (Γ)

$$\Gamma = \{(a_i, \alpha_i), i = 1, \dots, m\}$$

Knowledge base (Γ_α)

The knowledge base Γ is divided into nested level cuts:

$$\Gamma_\alpha = \{(a_i, \alpha_i) \in \Gamma \mid \alpha_i \geq \alpha\}$$

- Nested structure: $\Gamma_\alpha \subseteq \Gamma_\beta \quad \text{if } \beta \leq \alpha$.

Inference

$$\Gamma \vdash (a, \alpha) \iff \Gamma_\alpha \vdash (a, \alpha)$$

- A possibilistic inference (a, α) depends only on formulas with certainty levels $\geq \alpha$.

Example

- Knowledge Base:

$$\Gamma = \{(a_1, 0.9), (a_2, 0.7), (a_3, 0.5)\}.$$

- Inference for $\alpha = 0.7$:

$$\Gamma_\alpha \vdash (a_2, 0.7) \implies \Gamma \vdash (a_2, 0.7).$$

Inconsistency level of Γ

$$\text{inc-l}(\Gamma) = \max\{\alpha \mid \Gamma \vdash (\bot, \alpha)\}.$$

Properties

Formulas with certainty levels $>$ inc-l(Γ) are safe from inconsistency.

$$\text{inc-l}(\{(a_i, \alpha_i) \mid (a_i, \alpha_i) \in \Gamma \text{and } \alpha_i > \text{inc-l}(\Gamma)\}) = 0$$

Overview

Default rule

$$\Pi(a \wedge b) > \Pi(a \wedge \neg b)$$

"If a, then b is more plausible than $\neg b$."

Comparative possibility relation

$$a \wedge b >_{\Pi} a \wedge \neg b$$

Penguin Example: Rules and Constraints

Default Rules

$$d_1 : b \rightarrow f \quad (\text{"Birds fly"}),$$
$$d_2 : p \rightarrow \neg f \quad (\text{"Penguins do not fly"}),$$
$$d_3 : p \rightarrow b \quad (\text{"Penguins are birds"}).$$

- Constraints:

$$C_1 : b \wedge f >_\Pi b \wedge \neg f \quad (\text{birds fly}).$$
$$C_2 : p \wedge \neg f >_\Pi p \wedge f \quad (\text{penguins don't fly}).$$
$$C_3 : p \wedge b >_\Pi p \wedge \neg b \quad (\text{penguins are birds}).$$

Worlds (Ω)

Ω is the finite set of interpretations of the considered propositional language.

Worlds (Ω)

Worlds (Ω):

$$\Omega = \{\omega_0, \omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6, \omega_7\},$$

where each ω_i represents a unique combination of b, f, p (bird, fly, penguin):

$$\begin{aligned}
&\omega_0 : \neg b \wedge \neg f \wedge \neg p, \quad \omega_1 : \neg b \wedge \neg f \wedge p,\\
&\omega_2 : \neg b \wedge f \wedge \neg p, \quad \omega_3 : \neg b \wedge f \wedge p,\\
&\omega_4 : b \wedge \neg f \wedge \neg p, \quad \omega_5 : b \wedge \neg f \wedge p,\\
&\omega_6 : b \wedge f \wedge \neg p, \quad \omega_7 : b \wedge f \wedge p
\end{aligned}$$

Penguin Example: Updated Constraints

- Constraints:

$$\begin{aligned}
C_1 &: \max(\pi(\omega_6), \pi(\omega_7)) > \max(\pi(\omega_4), \pi(\omega_5)), \\
C_2 &: \max(\pi(\omega_5), \pi(\omega_1)) > \max(\pi(\omega_3), \pi(\omega_7)), \\
C_3 &: \max(\pi(\omega_5), \pi(\omega_7)) > \max(\pi(\omega_1), \pi(\omega_3)).
\end{aligned}$$

Partition

Worlds are ranked based on constraints:

$$\{\omega_0, \omega_2, \omega_6\} >_\Pi \{\omega_4, \omega_5\} >_\Pi \{\omega_1, \omega_3, \omega_7\}.$$

Possibility Levels

$$\begin{aligned}
\pi(\omega_0) = \pi(\omega_2) = \pi(\omega_6) = 1, \\
\pi(\omega_4) = \pi(\omega_5) = 2/3, \\
\pi(\omega_1) = \pi(\omega_3) = \pi(\omega_7) = 1/3
\end{aligned}$$

Penguin Example: Computing Necessity Measures

Necessity Measure ($N(A)$)

$$N(A) = \inf_{u \notin A} (1 - \pi(u)).$$

Computation

$$\begin{aligned}
N(\neg p \vee \neg f) &= \min\{1 - \pi(\omega_3), 1 - \pi(\omega_7)\} = \min\{1 - 1/3, 1 - 1/3\} = 2/3 \\
N(\neg b \vee f) &= \min\{1 - \pi(\omega_4), 1 - \pi(\omega_5)\} = \min\{1 - 2/3, 1 - 2/3\} = 1/3 \\
N(\neg p \vee b) &= \min\{1 - \pi(\omega_1), 1 - \pi(\omega_3)\} = \min\{1 - 1/3, 1 - 1/3\} = 2/3
\end{aligned}$$

Penguin Example: Resulting Possibilistic Logic Base

Resulting Possibilistic Logic Base (K)

$$K = \{(\neg p \vee \neg f, 2/3), (\neg p \vee b, 2/3), (\neg b \vee f, 1/3)\}.$$

Initial Information (Tweety is a Bird):

$$F = \{(b, 1)\} \quad \text{(Tweety is a bird with certainty 1).}$$

- From the rule $(\neg b \vee f, 1/3)$ in K, we know:

$$(\neg b \vee f, 1/3) \quad \text{and} \quad (b, 1) \vdash (f, \min(1, 1/3)) = (f, 1/3).$$

- $K \cup \{(b, 1)\} \vdash (f, 1/3)$
- Conclusion: If Tweety is a bird, we conclude that Tweety flies with certainty $\frac{1}{3}$.

Penguin Example: Reasoning with New Information

New Information (Tweety is a Penguin):

$$F = \{(b, 1), (p, 1)\} \quad \text{(Tweety is a bird and a penguin with certainty 1).}$$

- From the rule $(\neg p \vee \neg f, 2/3)$ in K, we know:

$$(\neg p \vee \neg f, 2/3) \quad \text{and} \quad (p, 1) \vdash (\neg f, \min(1, 2/3)) = (\neg f, 2/3).$$

- From the rule $(\neg b \vee f, 1/3)$ in K and the new information $(p, 1)$, we know :

$$(\neg b \vee f, 1/3) \quad \text{and} \quad (b, 1) \vdash (f, \min(1, 1/3)) = (f, 1/3).$$

- These results conflict because:

$$(\neg f, 2/3) \quad \text{and} \quad (f, 1/3) \vdash (\perp, \min(2/3, 1/3)) = (\perp, 1/3).$$

- $K \cup F \vdash (\perp, 1/3)$
- Conclusion: The knowledge base becomes inconsistent with certainty $\frac{1}{3}$.

Penguin Example: Reasoning Final Conclusion

Final Conclusion (Tweety is a Penguin):

- Despite inconsistency, we can still infer:

$$K \cup F \vdash (\neg f, 2/3).$$

$$(\neg p \vee \neg f, 2/3) \quad \text{and} \quad (p, 1) \vdash (\neg f, \min(1, 2/3)) = (\neg f, 2/3).$$

- Conclusion: Since Tweety is a penguin, we conclude that Tweety does not fly with certainty $\frac{2}{3}$.

Overview

Conclusion

- Possibilistic logic provides a robust framework for reasoning under uncertainty.
- Key features:
  - Certainty levels for qualitative inference.
  - Compatibility with classical logic.
  - Handling of conflicting information.
- Applications:
  - Medical diagnosis, decision-making, and knowledge representation.

Bibliography I

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Specificity of a Possibility Distribution

A distribution π is **at least as specific** as π' if:

$$\pi(u) \leq \pi'(u), \quad \forall u \in U.$$

Example

- **Complete knowledge:** $\pi(u_0) = 1$, $\pi(u) = 0$ for all $u \neq u_0$.
- **Complete ignorance:** $\pi(u) = 1$ for all $u \in U$.

Principle of minimal specificity

It states that any hypothesis not known to be impossible cannot be ruled out. It is a minimal commitment, cautious, information principle.

Key Properties of Measures

Duality Relationship

$$N(A) = 1 - \Pi(A^c).$$

Where A^c is the complement of A.

Possibility: Maxitivity Property

$$\Pi(A \cup B) = \max(\Pi(A), \Pi(B)).$$

Necessity: Minimization Property

$$N(A \cap B) = \min(N(A), N(B)).$$

- Certainty-qualified information: "A is certain to degree α" is modeled by the constraint:

$$N(A) \geq \alpha$$

- Least Specific possibility distribution:

$$\pi_{(A,\alpha)}(u) = \begin{cases} 1 & \text{if } u \in A, \\ 1-\alpha & \text{otherwise.} \end{cases}$$

Example

- If $\alpha = 1$: $\pi(A, \alpha)$ becomes the characteristic function of A.
- If $\alpha = 0$: $\pi(A, \alpha)$ represents total ignorance.

Example

- A doctor is trying to diagnose a patient based on symptoms and diseases:
  - $U = \{\text{flu}, \text{cold}, \text{allergy}\}$.
  - Possibility distribution π based on symptom severity:

$$\pi(\text{flu}) = 0.9, \quad \pi(\text{cold}) = 0.6, \quad \pi(\text{allergy}) = 0.3.$$

- The doctor evaluates the possibility and necessity measures for a hypothesis A:

$$A = \{\text{flu}, \text{cold}\}$$

- Calculation:

$$\Pi(A) = \sup_{u \in A} \pi(u) = \max(\pi(\text{flu}), \pi(\text{cold})) = 0.9.$$

$$N(A) = \inf_{u \notin A} (1 - \pi(u)) = 1 - \pi(\text{allergy}) = 1 - 0.3 = 0.7.$$

- Interpretation:
  - Possibility ($\Pi(A)$): Flu or cold is highly possible (0.9).
  - Necessity ($N(A)$): It is moderately certain (0.7) that the patient has either flu or cold.

Axioms and Inference Rules: Formula Weakening

Formula weakening

If $a \vdash b$ in classical logic, then $(a, \alpha) \vdash (b, \alpha)$.

Example

- Premise:

$$(a, 0.8) \quad (\text{"It is raining with certainty 0.8"}).$$

- Conclusion:

$$(b, 0.8) \quad (\text{"The ground is wet with certainty 0.8"}).$$

Penguin Example: Constraints and Plausibility Relationships

- Default Rule $d_1 : b \rightarrow f$ ("Birds fly"):
  - Makes worlds where birds fly ($b \wedge f$) more plausible than those where birds do not fly ($b \wedge \neg f$).
  - Affects:

$$\omega_6 \,\text{(bird flies)} \quad \text{vs.} \quad \omega_4, \omega_5 \,\text{(bird does not fly)}.$$

- Default Rule $d_2 : p \rightarrow \neg f$ ("Penguins do not fly"):
  - Makes worlds where penguins do not fly ($p \wedge \neg f$) more plausible than those where penguins fly ($p \wedge f$).
  - Affects:

$$\omega_5 \,\text{(penguin does not fly)} \quad \text{vs.} \quad \omega_7, \omega_3 \,\text{(penguin flies)}.$$

- Default Rule $d_3 : p \rightarrow b$ ("Penguins are birds"):
  - Ensures penguins are considered birds, influencing how penguin-related worlds (p) are ranked.

Applications of Basic Possibilistic Logic

- Default Reasoning
- Belief Revision and Information Fusion
- Decision under Uncertainty
- Applications in Diagnosis and Recognition